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TOOL AND METHOD FOR ONE-HANDED MANIPULATION AND PLACEMENT OF CONDUIT-JOINING SLEEVE

Abstract

A tool for one-handed operation includes an open-ended cylinder having two half-cylinders coupled at first circumferential ends thereof. A retainer, coupled to the open-ended cylinder, has a spring-bias to automatically couple the half-cylinders when second circumferential ends of the half-cylinders are in contact with one another. The retainer is operable by a single finger of a user's hand to overcome the spring-bias such that the half-cylinders may move relative to one another. A first axial end of the open-ended cylinder slidingly receives an open-ended sleeve, and a second axial end of the open-ended cylinder has an internal annular ledge that provides an annular seat for the open-ended sleeve. A set of apertures extend through radial walls of the open-ended cylinder with each aperture including a portion located a distance from the annular ledge that is greater than an axial length of the open-ended sleeve.

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Background/Summary

FIELD OF THE DISCLOSURE

[0001] This disclosure relates generally to tools used in the joining of conduits, and more particularly to tools and methods that support one-handed manipulation and placement of a rigid cylindrical sleeve used to join pneumatic lines or fuel lines such as those found onboard aircraft.

BACKGROUND

[0002] Pneumatic lines and fuel lines are used extensively throughout all types of aircraft. When ends of two pneumatic lines or ends of two fuel lines must be connected, it is common to employ what is known as a “Wiggins” fitting to create the necessary sealed connection between two lines. Since the integrity of pneumatic lines and fuel lines is critical for aircraft, proper Wiggins fitting installations are essential in aircraft maintenance. Briefly, a Wiggins fitting consists of an open-ended, rigid cylindrical sleeve and a cylindrical clamp. When properly positioned, the sleeve fits snugly over both ends of two lines that are to be connected and the clamp holds the sleeve in place.

[0003] During installation of a Wiggins fitting, the critical and tedious portion of the installation operation involves proper positioning of the sleeve without damaging the sleeve. The manipulation and proper placement of the sleeve can take multiple attempts spanning hours for five reasons. First, there is an extremely tight diametric mating tolerance between a sleeve and the portions of the two lines that the sleeve engages. Second, the sleeve's structural integrity must be pristine as any dimensional distortions, dents, and/or nicks in the sleeve can render the sleeve useless. Three, in order to properly position a sleeve to achieve the proper seal with the connected lines, an installation technician currently uses conventional tools (e.g., screwdrivers, hammers, wrenches, etc.) to apply a motive force to the sleeve during its manipulation and placement that frequently introduces unwanted dimensional distortions, dents, and/or nicks in the sleeve. Fourth, the locations of many of these line connecting operations are difficult for the installation technician to reach and/or see. The fifth and sometimes most problematic reason is that the locations of the lines requiring connections are often in cramped areas that only allow a technician to use a single hand to perform the connection task. As a result, the critical installation of a Wiggins-fitting sleeve is frequently difficult, time consuming, and frustrating.

SUMMARY

[0004] In accordance with methods and systems described herein, a tool includes an open-ended cylinder having a first half-cylinder and a second half-cylinder. Each of the first half-cylinder and second half-cylinder has a first circumferential end and a second circumferential end. The first circumferential end of the first half-cylinder is hingedly coupled to the first circumferential end of the second half-cylinder. A retainer is coupled to the open-ended cylinder. The retainer has a spring-bias to automatically couple the first half-cylinder to the second half-cylinder when the second circumferential end of the first half-cylinder is in contact with the second circumferential end of the second half-cylinder. The retainer is also adapted to be operable by a single finger of a user's hand to overcome the spring-bias wherein the first half-cylinder and second half-cylinder are then free to move relative to one another. A first axial end of the open-ended cylinder is adapted to slidingly receive an open-ended sleeve. A second axial end of the open-ended cylinder has an annular ledge at an internal region of the open-ended cylinder. The annular ledge is adapted to annularly seat against the open-ended sleeve so that the open-ended cylinder fully encases the open-ended sleeve. A set of apertures extend through radial walls of the open-ended cylinder. Each such aperture includes a portion located a distance from the annular ledge that is greater than an axial length of the open-ended sleeve.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] Other objects, features and advantages of the methods and systems described in the present disclosure will become apparent upon reference to the following description of the preferred embodiments and to the drawings, wherein corresponding reference characters indicate corresponding parts throughout the several views of the drawings and wherein:

[0006] FIG. **1** is a cross-sectional view of the ends of two conventional pneumatic lines or two conventional fuel lines that are to be connected and sealed by a Wiggins fitting;

[0007] FIG. **2** is a cross-sectional view of the ends of the two lines in FIG. **1** with a conventional Wiggins-fitting sleeve properly positioned thereon;

[0008] FIG. **3** illustrates a perspective view of one embodiment of a one-handed sleeve manipulator and placement tool in its closed position in accordance with various aspects as described herein;

[0009] FIG. **4** illustrates an end view of the tool taken along line **4-4** in FIG. **3** depicting the sleeve-receiving end of the tool in accordance with various aspects as described herein;

[0010] FIG. **5** illustrates an end view of the tool in its open position as viewed from the sleeve-retaining end of the tool in accordance with various aspects as described herein;

[0011] FIG. **6** illustrates a cross-sectional view of the tool taken along line **6-6** in FIG. **3** in accordance with various aspects as described herein;

[0012] FIG. **7** illustrates the cross-sectional view of the tool shown in FIG. **6** with an open-ended sleeve properly positioned in the tool in accordance with various aspects as described herein;

[0013] FIG. **8**A is a side view of a portion of the tool illustrating an embodiment of one tool retrieval and sleeve alignment aperture in accordance with various aspects as described herein;

[0014] FIG. **8**B is a side view of a portion of the tool illustrating another

[0015] embodiment of one tool retrieval and sleeve alignment aperture in accordance with various aspects as described herein;

[0016] FIG. **8**C is a side view of a portion of the tool illustrating another embodiment of one tool retrieval and sleeve alignment aperture in accordance with various aspects as described herein;

[0017] FIG. **8**D is a side view of a portion of the tool illustrating another embodiment of one tool retrieval and sleeve alignment aperture in accordance with various aspects as described herein;

[0018] FIG. **9** is a side view of a portion of the tool illustrating an embodiment of a tool retrieval and sleeve alignment aperture having sleeve alignment markings adjacent thereto in accordance with various aspects as described herein;

[0019] FIG. **10** is side view of a portion of the sleeve manipulator and placement tool illustrating the tool's latch disengaged from the tool's catch in accordance with various aspects as described herein;

[0020] FIG. **11**A is an isolated side view of the tool's catch and latch prior to their engagement in accordance with various aspects as described herein;

[0021] FIG. **11**B is an isolated side view of the tool's catch and latch at the commencement of their engagement in accordance with various aspects as described herein;

[0022] FIG. **11**C is an isolated side view of the tool's catch engaged by the latch in accordance with various aspects as described herein;

[0023] FIG. **11**D is an isolated side view of the tool's latch with a technician's disengagement force being applied thereto in accordance with various aspects as described herein; and

[0024] FIG. **12** is a cross-sectional view of another embodiment of a one-handed sleeve manipulator and placement tool further including a cushion material disposed on the tool's annular ledge in accordance with various aspects as described herein.

DETAILED DESCRIPTION

[0025] Referring now to the drawings, simultaneous reference will be made to FIGS. **1** and **2** where cross-sectional views of the ends of two pneumatic conduits/lines or fuel conduits/lines of the type that are generally connected using a Wiggins fitting. As is known in the art, a Wiggins fitting includes a cylindrical sleeve and a cylindrical clamp. The cylindrical sleeve and cylindrical clamp are not part of the present invention. FIG. **1** illustrates two conventional cylindrical lines **100** and **110** prior to the joining thereof by a cylindrical sleeve of a Wiggins fitting. As is known in the art, lines **100** and **110** are typically in axial alignment with one another (as shown) prior to the installation of a Wiggins-fitting sleeve. FIG. **2** illustrates lines **100** and **110** after a Wiggins-fitting sleeve **200** (hereinafter referred to simply as open-ended sleeve **200** or sleeve **200**) has been properly manipulated and placed such that a portion of each line **100** and **110** is fitted in sleeve **200**. It is critical that sleeve **200** be properly positioned without incurring any damage.

[0026] As is known in the art, lines **100** and **110** are configured identically in terms of their end structure. That is, the outboard end of line **100** has two spaced-apart annular ferrules **102**/**104** with an O-ring **106** captured there between. Similarly, the outboard end of line **110** has two spaced-apart annular ferrules **112**/**114** with an O-ring **116** captured there between. Sleeve **200** has a smooth inside surface **202** that sealingly engages O-rings **106**/**116** when sleeve **200** is properly positioned over the ends of lines **100** and **110** as illustrated in FIG. **2**. The axial ends of sleeve **200** at inside surface **202** may typically be tapered radially outward at **204** to facilitate manipulation of sleeve **200** during a placement operation. Sleeve **200** is configured such that its axial length “L.sub.S” allows all annular ferrules **102**, **104**, **112** and **114** to be encased within sleeve **200** when sleeve **200** is properly positioned as shown in FIG. **2**. Outer surface **206** of sleeve **200** may have multiple annular grooves **208** that mate with portions of a conventional Wiggins clamp (not shown) once sleeve **200** is properly positioned as shown in FIG. **2**. Installation of a conventional Wiggins clamp on a properly positioned sleeve **200** is a straightforward and well-known operation that is not a limitation of the present invention.

[0027] As mentioned previously herein, the tight tolerances between lines **100**/**110** and sleeve **200** make the manipulation and placement of sleeve **200** a difficult task without applying some type of mechanical pressure or force to an axial end of sleeve **200** and/or an external radial face of sleeve **200**. Unfortunately, the need to maintain shape/size integrity of sleeve **200** to ensure a perfect seal with O-rings **106**/**116** makes it extremely difficult to apply non-damaging axial or radial forces to sleeve **200** using conventional aircraft maintenance tools such as screwdrivers, hammers, wrenches, etc. These issues are further exacerbated in cases where the sleeve-placing technician only has room to utilize one hand to manipulate and properly place sleeve **200**, and then remove/retrieve whatever tool may have been used to aid in the manipulation and placement of the sleeve.

[0028] The present invention is a tool that provides for one-handed manipulation of sleeve **200** to assure proper placement of sleeve **200**. In addition, the tool protects and guarantees shape/size integrity of sleeve **200**, and drastically reduces the time required for proper/safe manipulation/placement of sleeve **200** as well as the subsequent removal and retrieval of the tool. While the present disclosure assumes that sleeve **200** is a Wiggins-fitting sleeve, it is to be understood that the tool described herein may be used to facilitate one-handed manipulation and placement of any type of open-ended sleeve.

[0029] Referring now simultaneously to FIG. **3-6**, a one-handed sleeve manipulation tool in accordance with an embodiment of the present disclosure is shown in a perspective view (FIG. **3**), in two axial end views thereof (i.e., tool is closed in FIG. **4** and open in FIG. **5**), and in a cross-sectional view thereof (FIG. **6**). The tool is referenced generally by numeral **10**. It is to be understood that some features of tool **10** to be described herein will not be visible in all views thereof. Tool **10** may be made of a rigid material(s) typically used in the manufacture of machine tools and may include stainless steel.

[0030] Tool **10** has two sections **12** and **14**, each of which is a half-cylinder as best seen in FIG. **5**. One circumferential end **12**A of section **12** is hingedly coupled to one circumferential end **14**A of section **14** at a hinge joint **16** that can be configured as a finger joint hinge. As would be understood in the art, a hinge pin **18** (FIGS. **3-5**) passes through finger joint hinge **16** such that sections **12**/**14** can rotate about hinge pin **18**. In some embodiments, finger joint hinge **16** may be configured such that sections **12** and **14** can experience relative rotation of at least 90° as indicated by two-headed arrow **20** (FIG. **5**) such that each of half-cylinder sections **12** and **14** is unencumbered by the other half-cylinder section when tool **10** is fully opened as shown in FIG. **5**. In this way, either half-cylinder section **12** or **14** may be readily positioned about a Wiggins fitting sleeve as will be explained further below.

[0031] When in its closed position as shown in FIGS. **3**, **4** and **6**, one open axial end **40** of tool **10** has a diameter "D.sub.1" sized to slidingly receive a Wiggins-fitting sleeve such as the above-described sleeve **200**. As will be explained further below, the entirety of a Wiggins-fitting sleeve fits axially within tool **10** such that the sleeve is encased by tool **10**. The other open axial end **42** of tool **10** in its closed position is configured to define an interior annular ledge **44** that extends partially and radially into tool **10** to form a stop for the sleeve fitted in tool **10** thereby preventing an encased sleeve from exiting open axial end **42**. The sleeve is properly positioned in tool **10** when the sleeve annularly seats against annular ledge **44**. The interior axial length "L" (FIG. **6**) of tool **10** measured from open axial end **40** to interior annular ledge **44** is greater than the axial length L.sub.S (FIG. **2**) of a sleeve that tool **10** will be used to manipulate and position thereby ensuring that the sleeve is protected.

[0032] When tool **10** is in its closed position, the inside diameter "D.sub.2" defined by annular ledge **44** (FIG. **4**) is larger than the outside diameter of a line it will be disposed about, but smaller than the outside diameter of the pneumatic or fuel lines' annular ferrules (e.g., previously-described annular ferrules **102**/**104** or **112**/**114** and as shown in FIGS. **1-2**). The axial external end face of end **42** (formed when sections **12** and **14** are coupled together at circumferential ends **12**B and **14**B, respectively) may include indentations or be knurled as indicated by reference numeral **46** for non-slip cooperation with a technician's hand or with a conventional or specialized tool used to apply force to axial end **42** as will be explained further below.

[0033] As mentioned above, an open-ended sleeve (e.g., sleeve **200** shown in FIG. **2**) that is to be placed by tool **10** is properly positioned in the tool when the sleeve is annularly seated against the tool's annular ledge **44**. Since the sleeve will be fully encased within tool **10**, it may be difficult for a technician to tell if the sleeve is properly seated in tool **10**. That is, because tool **10** is sized to slidingly but snugly receive an open-ended sleeve, it is possible for the sleeve to be canted and/or wedged at an angle in the tool such that the sleeve does not fully seat against annular ledge **44**. In addition, the close quarters where tool **10** is to be used makes the retrieval of a dropped tool very difficult. To alleviate these problems, tool **10** has a set of tool retrieval and sleeve alignment apertures **50** that extend through the radial walls of tool **10** that will be explained with simultaneous reference will be made to FIGS. **3**, **6** and **7**.

[0034] The apertures' tool retrieval features allow tool **10** to be easily retrieved using some type of hooked tool (not shown) if/when the tool is dropped in a cramped work environment. Briefly, each of apertures **50** includes a longer portion **51** and a shorter portion **52** that is contiguous with portion **51**. Longer portion **51** provides a technician with a relatively large region to receive a hooked tool (not shown), while shorter portion **52** provides a capture area for a hooked tool to settle in as the tool is being retrieved using the hooked tool.

[0035] In terms of the apertures' sleeve alignment features, portion **52** has an outboard end **52**A located an axial distance "Y" (FIG. **6**) from annular ledge **44** where distance Y is greater than the axial length Ls of the sleeve (e.g., sleeve **200** shown in FIGS. **2** and **7**) such that one axial end of sleeve **200** is visible in portions **52** of apertures **50** when the other axial end of sleeve **200** is seated against annular ledge **44** as shown in FIG. **7**. In this way, a technician can readily see that the sleeve is properly seated in the tool.

[0036] In the illustrated example of aperture **50** which is also shown in isolation in FIG. **8**A, aperture **50** has a first portion **51** that extends parallel to open axial end **40** and a second portion **52** that is contiguous with first portion **51** and perpendicular to open axial end **40**. It is to be understood that apertures **50** are not limited to the “L” or “J” shape shown in FIG. **8**A. For example, in some embodiments, aperture **50** may be configured in a “T” shape as illustrated in FIG. **8**B where portion **52** is still contiguous with portion **51** and is still perpendicular to open axial end **40**. In some embodiments, portion **51** of an aperture **50** may be canted at an acute angle relative to open axial end **40** as shown in FIG. **8**C, or at an obtuse angle relative to open axial end **40** as shown in FIG. **8**D. In some embodiments, all of apertures **50** for a tool may be the same. In some embodiments, a tool's apertures **50** may include a mixture of different configurations. However, in all cases, each aperture **50** will include a portion that is located an axial distance (i.e., distance Y shown in FIG. **6**) from the tool's annular ledge **44** that is greater than the axial length L.sub.S (FIG. **2**) of the sleeve that will be encased by the tool.

[0037] In some embodiments and as illustrated in FIG. **9**, a set of markings **54** may be provided on tool **10**. Each marking **54** is adjacent to portion **52** of an aperture **50** to provide an indication of where the axial end of a sleeve should be when the sleeve is fully seated against the tool's annular ledge as described above. That is, each marking **54** is located an axial distance (“L.sub.M”) from the tool's annular ledge **44** where axial distance L.sub.M is equal to the axial length L.sub.S (FIG. **2**) of the sleeve that is to be encased by the tool. Markings **54** may be two-dimensional and/or three-dimensional without departing from the scope of the present disclosure. Markings may be placed on the outside of the tool, on the inside of the tool, or on the inside and outside of the tool.

[0038] In general, one-handed use of tool **10** is made possible by a spring-biased retainer that engages and keeps tool **10** in its closed state when circumferential ends **12**B/**14**B are in contact with one another, and that may be operated/opened by a single finger of a technician's hand. That is, the retainer automatically couples half-cylinders **12**/**14** when circumferential ends **12**B/**14**B are brought into contact, and allows a technician to hold the tool while simultaneously uncoupling the half-cylinders at circumferential ends **12**B/**14**B. Once the half-cylinders are uncoupled, half-cylinders **12**/**14** are free to move/rotate relative to one another about hinge **16**.

[0039] In the illustrated embodiment, the retainer is referenced generally by numeral **60** in FIG. **3**, is shown in greater detail in FIG. **10**, and is shown in isolated views thereof in FIGS. **11**A-**11**D to illustrate the retainer's operational states. Not all features of retainer **60** are visible in all figures. Accordingly, the following description will make simultaneous reference to FIGS. **3**, **10**, and **11**A-**11**D.

[0040] Retainer **60** includes a catch bar **62** and a spring-loaded latch **64**. Catch bar **62** is coupled to half-cylinder **12** at its circumferential end **12**B as illustrated in FIG.

[0041] **10**. Catch bar **62** is positioned to extend axially along half-cylinder **12** with an exposed portion **62**A available for engagement by spring-loaded latch **64** when circumferential ends **12**B/**14**B are brought into contact with one another (FIG. **3**). Latch **64** includes a rigid latch arm **66** and a spring (“S” in FIGS. **11**A-**11**D) **68**. Latch arm **66** includes a beveled outboard end **66**A and a hook **66**B adjacent to outboard end **66**A. Spring **68** may be a torsion spring that couples the other end **66**C of latch arm **66** to circumferential end **14**B of half-cylinder **14** as best seen in FIG. **10**. Spring **68** biases latch arm **66** (to include hook **66**B) radially inward as controlled by the spring's biasing force as indicated by arrow **70** in FIG. **11**A. As circumferential ends **12**B/**14**B are brought close enough to one another such that beveled outboard end **66**A contacts catch bar **62** as shown in FIG. **11**B, the biasing force of spring **68** is exceeded so that latch arm **66** rotates radially outward about spring **68** as indicated by rotational arrow **74**. Once circumferential ends **12**B/**14**B contact one another, hook **66**B is aligned with catch bar **62** such that bias force **70** is again active to rotate latch arm **66** about spring **68** so that hook **66**B engages catch bar **62** as shown in FIG. **11**C. To disengage hook **66**B from catch bar **62**, a technician need only push on beveled end **66**A (with a force indicated by arrow **76** in FIG. **11**D) using a single finger (not shown). Once latch arm **66** is disengaged from catch bar **62**, the tool's two half-cylinders are free to move/rotate relative to one another as described above.

[0042] To use tool **10**, a sleeve (e.g., the above-described sleeve **200**) is positioned on a first line that is to be joined but away from its ferrule/O-ring end. For example, the first line's O-ring can be temporarily removed thereby allowing the sleeve to be readily slid over the line's annular ferrules. The O-ring is then re-installed between the first line's annular ferrules. Using one hand, a technician then places tool **10** near where the connection task is to be performed. Using that same hand, the technician opens tool **10** by disengaging retainer **60** as described above and places one of half-cylinder sections **12** or **14** about the sleeve with the tool's open axial end **40** located closest to or facing the first line's ferrule/O-ring end. Tool **10** is closed by the technician using the same hand where the tool's retainer **60** automatically coupled the tools' half-cylinders and maintains the tool in its closed position about the sleeve and line as described above. At this point, tool **10** completely encases and protects the sleeve. With the sleeve fully protected, the technician may verify that the sleeve is properly seated against the tool's annular ledge using apertures **50** as described above. Once the proper seating of the sleeve is verified, forces may be applied to the tool (e.g., at the axial external end face of axial end **42**) using the same hand with or without the aid of a conventional tool such as a flat-head screwdriver. Axial pressure may be applied to end **42** in increments at different circumferential locations on axial end **42**. The axial pressure applied to tool **10** is distributed evenly about the axial end of the sleeve by annular ledge **44**.
[0043] The manipulation of tool **10** along with the encased/protected sleeve proceeds over the first line's ferrule/O-ring end and, subsequently, over the ferrule/O-ring end of the to-be-joined second line. The inside diameter D.sub.2 defined by the interior of annular ledge **44** causes annular ledge **44** to engage with the first of two annular ferrules on the first line to prevent further axial movement of tool **10** and the sleeve encased thereby. When this occurs, the sleeve is properly positioned over both lines (as shown in FIG. **2**) and tool **10** can be opened and removed. That is, once the sleeve has been manipulated into its proper location, the technician may use one finger form the same hand that has been used throughout the process to quickly open retainer **60** as described above.
[0044] The advantages of the present invention are numerous. The above-described tool provides for the one-handed manipulation and correct placement of a Wiggins-fitting sleeve on two pneumatic lines or fuel lines when the lines are placed end-to-end, and then provide for the subsequent one-handed removal of the tool. The tool provides the means to drastically reduce the time it takes for an aircraft maintenance task that must be performed on a regular basis. The tool protects the Wiggins-fitting sleeve from damage to ensure the integrity of the ultimate line connection.
[0045] Although the methods and systems presented herein have been described for specific embodiments thereof, there are numerous variations and modifications that will be readily apparent to those skilled in the art in light of the above teachings. For example, in some embodiments and as shown in FIG. **12**, a cushion material **48** (e.g., rubber) can be disposed on interior annular ledge **44** for further protection of the axial end of a sleeve during an installation operation. It is therefore to be understood that, within the scope of the appended claims, the methods and systems presented herein may be practiced other than as specifically described.
[0046] What is claimed as new and desired to be secured by Letters Patent of the United States is:

Claims

1. A tool, comprising: an open-ended cylinder having a first half-cylinder and a second half-cylinder, each of said first half-cylinder and said second half-cylinder having a first circumferential end and a second circumferential end, wherein said first circumferential end of said first half-cylinder is hingedly coupled to said first circumferential end of said second half-cylinder; a retainer coupled to said open-ended cylinder, said retainer having a spring-bias for automatically coupling said first half-cylinder to said second half-cylinder when said second circumferential end of said

first half-cylinder is in contact with said second circumferential end of said second half-cylinder, said retainer adapted to be operable by a single finger of a user's hand to overcome said spring-bias wherein said first half-cylinder and said second half-cylinder are free to move relative to one another; a first axial end of said open-ended cylinder adapted to slidingly receive an open-ended sleeve; a second axial end of said open-ended cylinder having an annular ledge at an internal region of said open-ended cylinder, said annular ledge adapted to annularly seat against the open-ended sleeve wherein said open-ended cylinder fully encases the open-ended sleeve; and a set of apertures extending through radial walls of said open-ended cylinder, each aperture from said set of apertures having a first portion extending substantially along said first axial end and having a second portion contiguous with said first portion and perpendicular to said fist axial end, said second portion having an outboard end wherein a distance from said annular ledge to said outboard end is greater than an axial length of the open-ended sleeve.
2. The tool of claim 1, wherein said first portion is longer than said second portion.
3. The tool of claim 1, wherein said first portion is parallel to said first axial end.
4. The tool of claim 1, wherein an axial external end face of said second axial end of said open-ended cylinder is knurled.
5. The tool of claim 1, further comprising a cushion material disposed on said annular ledge.
6. The tool of claim 1, wherein said open-ended cylinder is adapted to be disposed about a conduit having an annular ferrule at the conduit's exterior surface, and wherein said annular ledge has an inside diameter that is greater than a diameter of the conduit and less than a diameter of the annular ferrule.
7. The tool of claim 1, further comprising: a set of markings on said open-ended cylinder, each marking from said set of markings adjacent to one said second portion, wherein a distance from said annular ledge to said each marking is equal to the axial length of the open-ended sleeve.
8. A tool, comprising: an open-ended cylinder having a first half-cylinder and a second half-cylinder, each of said first half-cylinder and said second half-cylinder having a first circumferential end and a second circumferential end, wherein said first circumferential end of said first half-cylinder is hingedly coupled to said first circumferential end of said second half-cylinder; a catch coupled to said second circumferential end of said first half-cylinder; a latch coupled to said second circumferential end of said second half-cylinder, said latch having a spring-bias for automatic engagement with said catch when said second circumferential end of said first half-cylinder is in contact with said second circumferential end of said second half-cylinder, said latch adapted to be operable by a single finger of a user's hand to overcome said spring-bias wherein said first half-cylinder and said second half-cylinder are free to move relative to one another; a first axial end of said open-ended cylinder adapted to slidingly receive an open-ended sleeve; a second axial end of said open-ended cylinder having an annular ledge at an internal region of said open-ended cylinder, said annular ledge adapted to annularly seat against the open-ended sleeve wherein said open-ended cylinder fully encases the open-ended sleeve; and a set of apertures extending through radial walls of said open-ended cylinder, each aperture from said set of apertures including a portion located a distance from said annular ledge that is greater than an axial length of the open-ended sleeve.
9. The tool of claim 8, wherein said catch comprises a bar extending axially along said open-ended cylinder, and wherein said latch includes a hook for engagement with said bar when said second circumferential end of said first half-cylinder is in contact with said second circumferential end of said second half-cylinder wherein contact is maintained between said second circumferential end of said first half-cylinder and said second circumferential end of said second half-cylinder.
10. The tool of claim 8, wherein an axial external end face of said second axial end of said open-ended cylinder is knurled.
11. The tool of claim 8, further comprising a cushion material disposed on said annular ledge.
12. The tool of claim 8, wherein said open-ended cylinder is adapted to be disposed about a conduit

having an annular ferrule at the conduit's exterior surface, and wherein said annular ledge has an inside diameter that is greater than a diameter of the conduit and less than a diameter of the annular ferrule.

13. The tool of claim 8, further comprising: a set of markings on said open-ended cylinder, each marking from said set of markings adjacent to one said portion, wherein a distance from said annular ledge to said each marking is equal to the axial length of the open-ended sleeve.

14. A tool, comprising: an open-ended cylinder having a first half-cylinder and a second half-cylinder, each of said first half-cylinder and said second half-cylinder having a first circumferential end and a second circumferential end, wherein said first circumferential end of said first half-cylinder is hingedly coupled to said first circumferential end of said second half-cylinder; a catch coupled to said second circumferential end of said first half-cylinder; a latch coupled to said second circumferential end of said second half-cylinder, said latch having a spring-bias for automatic engagement with said catch when said second circumferential end of said first half-cylinder is in contact with said second circumferential end of said second half-cylinder, said latch adapted to be operable by a single finger of a user's hand to overcome said spring-bias wherein said first half-cylinder and said second half-cylinder are free to move relative to one another; a first axial end of said open-ended cylinder adapted to slidingly receive an open-ended sleeve; a second axial end of said open-ended cylinder having an annular ledge at an internal region of said open-ended cylinder, said annular ledge adapted to annularly seat against the open-ended sleeve wherein said open-ended cylinder fully encases the open-ended sleeve; and a set of apertures extending through radial walls of said open-ended cylinder, each aperture from said set of apertures having a first portion extending substantially along said first axial end and having a second portion contiguous with said first portion and perpendicular to said fist axial end, said second portion having an outboard end wherein a distance from said annular ledge to said outboard end is greater than an axial length of the open-ended sleeve.

15. The tool of claim 14, wherein said first portion is longer than said second portion.

16. The tool of claim 14, wherein said first portion is parallel to said first axial end.

17. The tool of claim 14, wherein an axial external end face of said second axial end of said open-ended cylinder is knurled.

18. The tool of claim 14, further comprising a cushion material disposed on said annular ledge.

19. The tool of claim 14, wherein said open-ended cylinder is adapted to be disposed about a conduit having an annular ferrule at the conduit's exterior surface, and wherein said annular ledge has an inside diameter that is greater than a diameter of the conduit and less than a diameter of the annular ferrule.

20. The tool of claim 14, further comprising: a set of markings on said open-ended cylinder, each marking from said set of markings adjacent to one said second portion, wherein a distance from said annular ledge to said each marking is equal to the axial length of the open-ended sleeve.

21. The tool of claim 14, wherein said catch comprises a bar extending axially along said open-ended cylinder, and wherein said latch includes a hook for engagement with said bar when said second circumferential end of said first half-cylinder is in contact with said second circumferential end of said second half-cylinder wherein contact is maintained between said second circumferential end of said first half-cylinder and said second circumferential end of said second half-cylinder.

22. A tool, comprising: an open-ended cylinder having a first half-cylinder and a second half-cylinder, each of said first half-cylinder and said second half-cylinder having a first circumferential end and a second circumferential end, wherein said first circumferential end of said first half-cylinder is hingedly coupled to said first circumferential end of said second half-cylinder; a retainer coupled to said open-ended cylinder, said retainer having a spring-bias for automatically coupling said first half-cylinder to said second half-cylinder when said second circumferential end of said first half-cylinder is in contact with said second circumferential end of said second half-cylinder, said retainer adapted to be operable by a single finger of a user's hand to overcome said spring-bias wherein said first half-cylinder and said second half-cylinder are free to move relative to one another; a first axial end of said open-ended cylinder adapted to slidingly receive an open-ended sleeve; a second axial end of said open-ended cylinder having an annular ledge at an internal region of said open-ended cylinder, said annular ledge adapted to annularly seat against the open-ended sleeve wherein said open-ended cylinder fully encases the open-ended sleeve; and a set of apertures extending through radial walls of said open-ended cylinder, each aperture from said set of apertures including a portion located a distance from said annular ledge that is greater than an axial length of the open-ended sleeve.

23. The tool of claim 22, further comprising: a set of markings on said open-ended cylinder, each marking from said set of markings adjacent to one said portion, wherein a distance from said annular ledge to said each marking is equal to the axial length of the open-ended sleeve.

24. A method, comprising: by a tool having an open-ended cylinder with a first half-cylinder and a second half-cylinder, each of the first half-cylinder and second half-cylinder having a first circumferential end and a second circumferential end, where the first circumferential end of the first half-cylinder is hingedly coupled to the first circumferential end of the second half-cylinder, and where the open-ended cylinder has a first axial end and a second axial end with the second axial end having an annular ledge at an internal region of the open-ended cylinder, a retainer coupled to the open-ended cylinder and having a spring-bias for automatically coupling the first half-cylinder to the second half-cylinder when the second circumferential end of the first half-cylinder is in contact with the second circumferential end of the second half-cylinder, and a set of apertures extending through radial walls of the open-ended cylinder with each aperture including a portion located a distance from the annular ledge, positioning, by a single hand of a user, the open-ended cylinder about an open-ended sleeve disposed about a conduit, wherein the open-ended sleeve has an axial length that is less than the distance of the portion from the annular ledge; verifying, using at least one of the apertures, that the open-ended cylinder is annularly seated against annular ledge wherein the open-ended cylinder fully encases the open-ended sleeve; manipulating, by the single hand of the user, the open-ended cylinder with the encased open-ended sleeve along the conduit to a desired location; and operating, by a single finger of the single hand of the user, the retainer to overcome the spring-bias wherein the first half-cylinder and second half-cylinder are free to move relative to one another.

25. The method of claim 24, wherein the step of verifying comprises: comparing a position of the encased open-ended sleeve to a marking on the open-ended cylinder, wherein the marking is adjacent to the portion of an aperture and is located a marking distance from the annular ledge that is equal to the axial length of the open-ended sleeve.